2013 F=ma Exam: Problem 5

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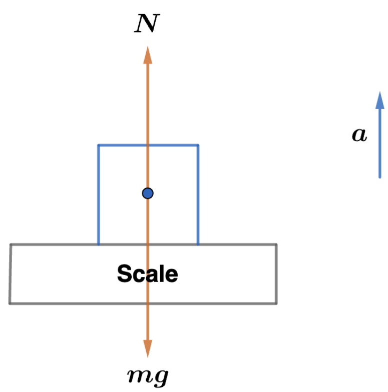
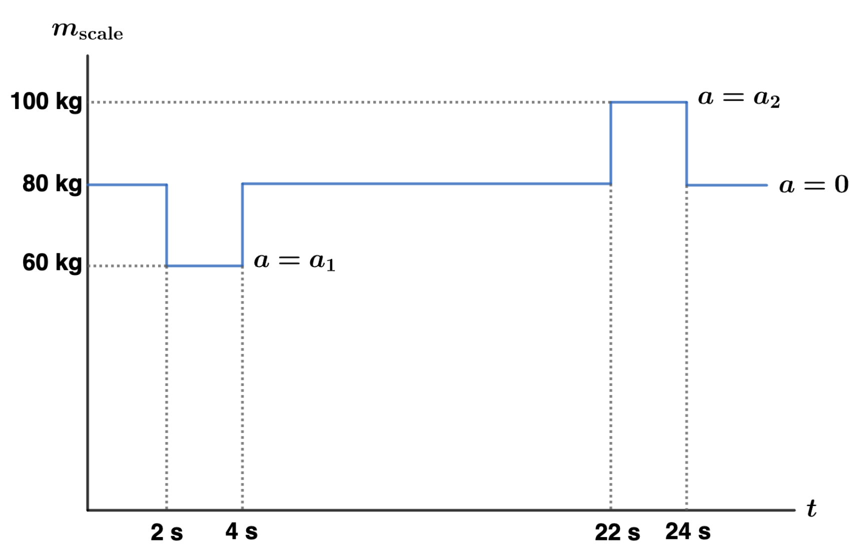


The scale reads the normal force N and assumes it is the weight mg. The measured mass is then $m_{\text{scale}} = N/g$. If the elevator has acceleration a upward, we have

$$N - mg = ma$$
$$N = m(g + a)$$

so

$$m_{\text{scale}} = \frac{N}{g} = m\left(1 + \frac{a}{g}\right)$$



In the beginning, the elevator is at rest so $a = 0$ and $m_{\text{scale}} = m = 80\,\text{kg}$. Once the elevator starts moving, the acceleration can be found from the measured mass,

$$a = g\left(\frac{m_{\text{scale}}}{m} - 1\right)$$

Thus, between $t = 2\,\text{s}$ and $4\,\text{s}$ we have acceleration

$$a_1 = (10\,\text{m/s}^2)\left(\frac{60\,\text{kg}}{80\,\text{kg}} - 1\right) = -2.5\,\text{m/s}^2$$

Between $t = 22\,\text{s}$ and $24\,\text{s}$ we have acceleration

$$a_2 = (10\,\text{m/s}^2)\left(\frac{100\,\text{kg}}{80\,\text{kg}} - 1\right) = 2.5\,\text{m/s}^2$$

The student has maximum downward velocity after the downward acceleration of the elevator but before it slows down with upward acceleration. This is from $t = 4\,\text{s}$ to $22\,\text{s}$ so the answer is $\boxed{\text{C}}$.